

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                    | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)   |  | High frequency / high energy / short wavelength (1)<br>electromagnetic wave (1)                                                                                                                                                                                                                    | 2               |     |     | 2     |       |      |
|          |     | (ii)  |  | Easily pass through the body / highly penetrative(1)<br>with low ionising power (1)                                                                                                                                                                                                                | 2               |     |     | 2     |       |      |
|          |     | (iii) |  | Thallium–(201) (1)<br>because it will <u>decay quicker</u> than the others / won't be exposed<br>to radiation for too long(1)<br><br>Do not accept shortest half life                                                                                                                              |                 | 1   | 1   | 2     |       |      |
|          | (b) |       |  | $1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \frac{1}{16} \rightarrow \frac{1}{32}(1)$<br>= 5 half lives (1)<br>time = 5 (ecf) × 17 (ecf) (1)<br>ans = 85 days (1)<br><br>only allow ecf if first line arrives at an answer other than 5 and is<br>shown |                 | 4   |     | 4     | 3     |      |
|          |     |       |  | Question 5 total                                                                                                                                                                                                                                                                                   | 4               | 5   | 1   | 10    | 3     | 0    |